Coursework – Canadian Households Housing Burden

This coursework is focused on studying Canadian households and their household disposable income spent on shelter expenditures. For this, we have been exclusively provided with the data from Environics Analytics, a leading company that has "Canada's largest collection of high-quality, actionable data". We have been provided with an exclusive license to use this data for the course. Please note that **THIS DATA CANNOT BE SHARED AND MUST BE DELETED AFTER FINISHING THE COURSE**. Please read the data disclosure agreement at the end of the coursework description.

You have been provided with two datasets, covering most Canadian households:

- Household spending per dissemination area.
- Demographics data per dissemination area (PRCDDA) level.

The documentation of each dataset can be seen in the Documentation folder of each dataset.

With this information, write a **business consultancy report** [1]with your analysis of the data, answering the following questions. Note that the code is NOT the core of the problem, the insights, analyses, and how you interpret the data is.

1. **(10%) Data preparation.** To be able to solve the coursework, we will first combine the available datasets into a unique tableau.
    a. The raw CSVs contain several geographic levels in the GEO column (province, CMA, CSD, PRCDDA, etc.). Filter your data to focus on PRCDDA (Dissemination Area).
    b. Construct the target variable. Define the Housing Burden Ratio (HBR) per PRCDDA as.

    $$\boldsymbol{y = \frac{HSSH001S}{HSAGDISPIN}}$$

    Since y is derived from shelter expenditure and disposable income, you must remove all variables mathematically related to either. Using shelter sub-items to predict a shelter-based ratio is circular. To avoid leakage, remove all variables starting with HSSH (e.g. HSSH001, HSSH001S, HSSH022, HSSH034B, etc.) and remove the aggregates: HSAGDISPIN, HSAGDISCIN, HSHNIAGG, HSTT001, HSTE001, HSTX001, HSTC001.

    Note: If you reduce features using any method that involves y, do so on the training set only.

    c. Construct the tableau. Construct the variables that can be predictive for the above target. As we only have cross-sectional data, our analysis will not be fully predictive,

[1] Guides such as https://slidemodel.com/consulting-report-how-to-write-and-present-one/ can help.

but it will be diagnostic of what are the main points that determine a PRCDDA household expenditure.

2. **(40%) Unsupervised Learning.**
   The first part of the coursework will focus on identifying the characteristics of Canadian households, excluding shelter spending. For this, do not include, in your clustering and dimensionality reduction models, the target you just built nor any variables that present data leakage.
   a. (10%) Task 1.1
      i. Load and merge the datasets. Report the dimensions of the merged data.
      ii. Remove non-numeric columns (CODE, GEO), zero-variance columns, and handle missing values. Justify your choices.
      iii. After removing the variables with potential data leakage, report the total number of remaining features and standardise them.
      iv. Create a train test split of your data. Justify your choices. From here onwards, all models must be implemented over the train test, and applied over the test set, unless this is not possible. Always mention this if necessary.
      v. Examine the distribution of the target variable y. Is it skewed?
      vi. Identify and visualise the top predictors correlated with y (e.g. bar chart of absolute correlations). Provide at least one additional visualisation that reveals an interesting pattern in the data and discuss your results.

   b. (10%) **K-Means clustering**
      i. Run KMeans for k = 2,...,12.
      ii. Record the silhouette score for each k and plot the Elbow curve and Silhouette scores for each k.
      iii. Using your previous results, select the optimal k and justify. Do the two methods agree? Note: Different preprocessing (e.g. log transformation, feature subsets) can change the optimal k. Discuss how your choices may have influenced the result.
      iv. Report cluster sizes. Compare feature means on interpretable variables (e.g. income, house-hold size, immigration status) and give each cluster a descriptive name. Explain your reasoning and insights.
      v. Create a boxplot of the target variable y by cluster. Do the clusters capture meaningful differences in housing expenditure patterns?

   c. (10%) **PCA.**
      Now we will apply a linear dimensionality reductions technique to the data.
      i. Perform PCA. Report the cumulative variance explained by the top 10 components.
      ii. Construct a scatter plot of the PC1 vs PC2, coloured by the KMeans labels you obtained on the previous questions. Are the clusters clearly separated? Does the data satisfy the cluster assumption?

iii. For the top 3 PCs, list top-loading features and interpret what each component represents.
iv. Compute mean PC scores by cluster (PC1–PC3) and propose human-readable cluster names based now on the PCA concepts. How do they differ from your previous ones?

d. (10%) **UMAP.**
   i. Use UMAP to reduce the data to two dimensions. Justify your choice of parameters by searching for the optimal value as you deem reasonable.
   ii. As before, create a scatter plot of your resulting dimensions coloured by the KMeans labels. Is UMAP a better or worse representation than PCA?
   iii. Compute the silhouette score in 2D UMAP space and compare with the original high-dimensional score. What does this tell you about the UMAP representation and your data? Which clusters do you believe to be the most informative from all three sets you created?

3. (40%) **Supervised Learning.** Now we will create models for a household's housing burden ratio y.
   a. (15%) **Elastic-Net GLM.**
   Let's start by training a regularized elastic net linear regression from your data.
      i. As our data is bounded, we will train a Generalized Linear Model of our data. For this, we will first transform our data to a different space. The best distribution that fits the data is a Pearson Type III distribution (PT3). Fit a Pearson Type III (PT3) distribution to your target variable using the distfit package over the train set. Take your target value y, your fitted distribution, and calculate where on the CDF of the PT3 distribution your y fits, calling this y_{PT3}. Then, using the normal CDF inverse (PPF function), find the equivalent z_{Normal}. This will be your new target for this question. (*Hint: You may need to slightly bound or clip your CDF values away from exactly 0 and 1 to avoid infinite values when applying the normal PPF.*)
      ii. Apply any transformations to your input data you deem necessary for the model to correctly converge.
      iii. Train an ElasticNet Linear Regression via GridSearchCV (5-fold) using your z_{Normal} as the target. Search over at least 5 alpha values in $[10^{-4}, 1]$ and at least four l1_ratio values in [0, 1].
      iv. Reverse the process in 3.a.i to get your predictions in the y original space.
      v. Report the best hyperparameters, test R2, and test MSE. Compute a bootstrapped 95% CI for R2 (at least 1,000 iterations). Finally, calculate a scatterplot of your predicted values versus the target values. What do these results tell you about the performance of your model?
      vi. Calculate the R2 (with 95% CI) per K-Means label from 1.b. Are there any meaningful differences in the prediction between groups? Given your understanding of the groups you got, why would this happen?

vii. Report the number of non-zero coefficients. Interpret the top 5 by magnitude. Are they reasonable? What do they tell you about the target variable?

b. (15%) **XGBoost.** Now we will train an XGB model.
   i. Follow the same process for the previous question to prepare the data, if necessary. As XGB is a non-linear model, you do NOT need to change the space of y, you can use y as your target directly.
   ii. Train an XGBoost regressor via RandomizedSearchCV (5-fold). Search over at least: n_estimators, max_depth, learning_rate, subsample, colsample_bytree. Decide on which numbers to use for this and discuss your parameter grid choices.
   iii. Calculate the same scatterplot and performance measures as before. Compare the CIs of the model versus the generalized linear model. Is it better? Worse? Why do you think this is? Discuss your findings and what would you recommend to use for a decision maker trying to predict household expenditure.

c. (10%) **SHAP explainer.**
   i. Compute the SHAP values for the XGBoost model using shap.TreeExplainer. Interpret what you see in the context of the problem.
   ii. Generate a SHAP summary plot (beeswarm) and list the top 10 features by mean absolute SHAP. Interpret your results.
   iii. Create at least two SHAP dependence plots for important features that you consider interesting to study. Discuss your choices and your interpretation of the results.
   iv. What are the most important variables? How do they compare to ElasticNet's coefficients? Is this problem non-linear?
   v. Compare the housing burden distribution across your Part 1.b clusters. Do they differ meaningfully in y? Are the features driving housing burden the same ones separating clusters, or are they capturing different aspects of the data?

The remaining 10% of the coursework corresponds to the format and language evaluations.

Conditions of the coursework

Software: You must use Python to run the numerical calculations over the dataset. A copy of your Jupyter notebook must be attached to the coursework in the submission (submit both the ipynb file and your report in PDF format). The notebook text MUST be attached in ipynb format, otherwise **a 25% discount will apply**.

Reproducibility: Use seed=42 whenever possible to allow reproducibility.

Word Limit: 3000 words +/-10% either side of the word count is deemed to be acceptable. Any text that exceeds an additional 10% will not attract any marks. The relevant word count *includes* items such as cover page, executive summary, title page, table of contents, tables, figures, in-text citations and section headings if used. The relevant word count *excludes* your list of references and any appendices at the end of your coursework submission (including the code).

You should always include the word count (from Microsoft Word, not Turnitin), at the end of your coursework submission, before your list of references.

Title/Cover Page: You must include a title/cover page that includes: your group's Student IDs and names, Course Code, Assignment Title, Word Count.

Submission Deadline: April 10th, 23:59.

Turnitin Submission: The assignment MUST be submitted electronically via Brightspace OWL. All required papers may be subject to submission for textual similarity review to the commercial plagiarism detection software under license to the University for the detection of plagiarism. All papers submitted for such checking will be included as source documents in the reference database to detect plagiarism of papers subsequently submitted to the system. Use of the service is subject to the licensing agreement, currently between The University of Western Ontario and Turnitin.com (http://www.turnitin.com).

AI Policy. AI is permitted only for code generation, documentation, and proofing only. Text-wise, it can be used for grammar and spelling checking. NO AI GENERATED TEXT WILL BE ALLOWED. Your insights must come from the group. If we have doubts, we may call the group for a presentation on their results where they will need to explain every insight and how they got to them.

IMPORTANT DATA USAGE NOTICE

Environics Analytics (EA) (https://environicsanalytics.com/about) has kindly provided this dataset for use within Data Science 3000B - Introduction to Machine Learning. The dataset provided by Environics Analytics for this coursework is confidential and protected under a Non-Disclosure Agreement (NDA) signed between EA and UWO. By accessing this data, you acknowledge and agree to the following conditions:

1. The data is watermarked and can only be used for coursework purposes within Data Science 3000B - Introduction to Machine Learning.
2. Sharing, distributing, or using this data outside of DS 3000B coursework is strictly prohibited.
3. All copies of the data must be permanently deleted upon completion of the course.
4. Any unauthorized use, sharing, or retention of this data may result in legal action.

By proceeding with this coursework, you acknowledge these terms and agree to comply with all data usage restrictions.